

Level 2 Practice Programs

Java program solutions without comments

Each question is shown with a clean, compile-ready solution.

Question 1: Factors, Sum, Sum of Squares, and Product

```
import java.util.Arrays;
import java.util.Scanner;

public class FactorsProgram {
    public static int countFactors(int number) {
        int count = 0;
        for (int i = 1; i <= number; i++) {
            if (number % i == 0) {
                count++;
            }
        }
        return count;
    }

    public static int[] getFactors(int number) {
        int[] factors = new int[countFactors(number)];
        int index = 0;
        for (int i = 1; i <= number; i++) {
            if (number % i == 0) {
                factors[index++] = i;
            }
        }
        return factors;
    }

    public static long sumOfFactors(int[] factors) {
        long sum = 0;
        for (int factor : factors) {
            sum += factor;
        }
        return sum;
    }

    public static long productOfFactors(int[] factors) {
        long product = 1;
        for (int factor : factors) {
            product *= factor;
        }
        return product;
    }

    public static long sumOfSquaresOfFactors(int[] factors) {
        long sum = 0;
        for (int factor : factors) {
            sum += (long) Math.pow(factor, 2);
        }
        return sum;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
System.out.print("Enter a number: ");
int number = sc.nextInt();

int[] factors = getFactors(number);
System.out.println("Factors: " + Arrays.toString(factors));
System.out.println("Sum of factors: " + sumOfFactors(factors));
System.out.println("Product of factors: " + productOfFactors(factors));
System.out.println("Sum of squares of factors: " + sumOfSquaresOfFactors(factors));
sc.close();
    }
}
```

Question 2: Recursive Sum of Natural Numbers

```
import java.util.Scanner;

public class RecursiveSumProgram {
    public static long recursiveSum(long n) {
        if (n == 1) {
            return 1;
        }
        return n + recursiveSum(n - 1);
    }

    public static long formulaSum(long n) {
        return n * (n + 1) / 2;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a natural number: ");
        long n = sc.nextLong();

        if (n <= 0) {
            System.out.println("Invalid input");
        } else {
            long sum1 = recursiveSum(n);
            long sum2 = formulaSum(n);
            System.out.println("Sum using recursion: " + sum1);
            System.out.println("Sum using formula: " + sum2);
            System.out.println("Both results are equal: " + (sum1 == sum2));
        }
        sc.close();
    }
}
```

Question 3: Leap Year

```
import java.util.Scanner;

public class LeapYearProgram {
    public static boolean isLeapYear(int year) {
        return (year % 4 == 0 && year % 100 != 0) || (year % 400 == 0);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter year: ");
        int year = sc.nextInt();

        if (year < 1582) {
            System.out.println("Year should be 1582 or later");
        } else if (isLeapYear(year)) {
            System.out.println(year + " is a Leap Year");
        } else {
            System.out.println(year + " is not a Leap Year");
        }
        sc.close();
    }
}
```